

# PERSONA: Psychological Profile Construction and Behaviorally Faithful Review Simulation via Multi-Signal Behavioral Twins

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DSN × BCT LLM Agent Challenge — Task A: User Modeling

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## Abstract

We present **PERSONA**, a user modeling system that constructs rich psychological profiles from review histories and uses them to simulate behaviorally faithful star ratings and written reviews for unseen items. Rather than treating users as static preference vectors, PERSONA builds a five-layer behavioral twin — encoding rating calibration, vocabulary fingerprint, value priority graph, cultural register, and temporal trajectory — and uses these layers jointly during simulation. On the Yelp dataset, PERSONA achieves a rating RMSE of **0.72** against a mean-rating baseline of **1.14** (37% reduction) and a corpus ROUGE-L of **0.31** for generated text. For Nigerian-English users, cultural contextualization yields pidgin-inflected reviews scoring 87/100 on Cultural Accuracy. The system is fully containerized, production-deployed over HTTPS, and includes a cold-start module that bootstraps a working profile from four conversational questions.

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## 1. Introduction

Every review is simultaneously a preference declaration, a stylistic fingerprint, and a cultural artifact. Yet most user modeling systems reduce this richness to a single latent embedding or a mean rating — discarding tone, habit, and context.

PERSONA answers a more ambitious question: can an agent understand a user deeply enough to impersonate them? Not just predict a rating, but write a review that sounds like that person, carries their biases, and reflects their cultural voice.

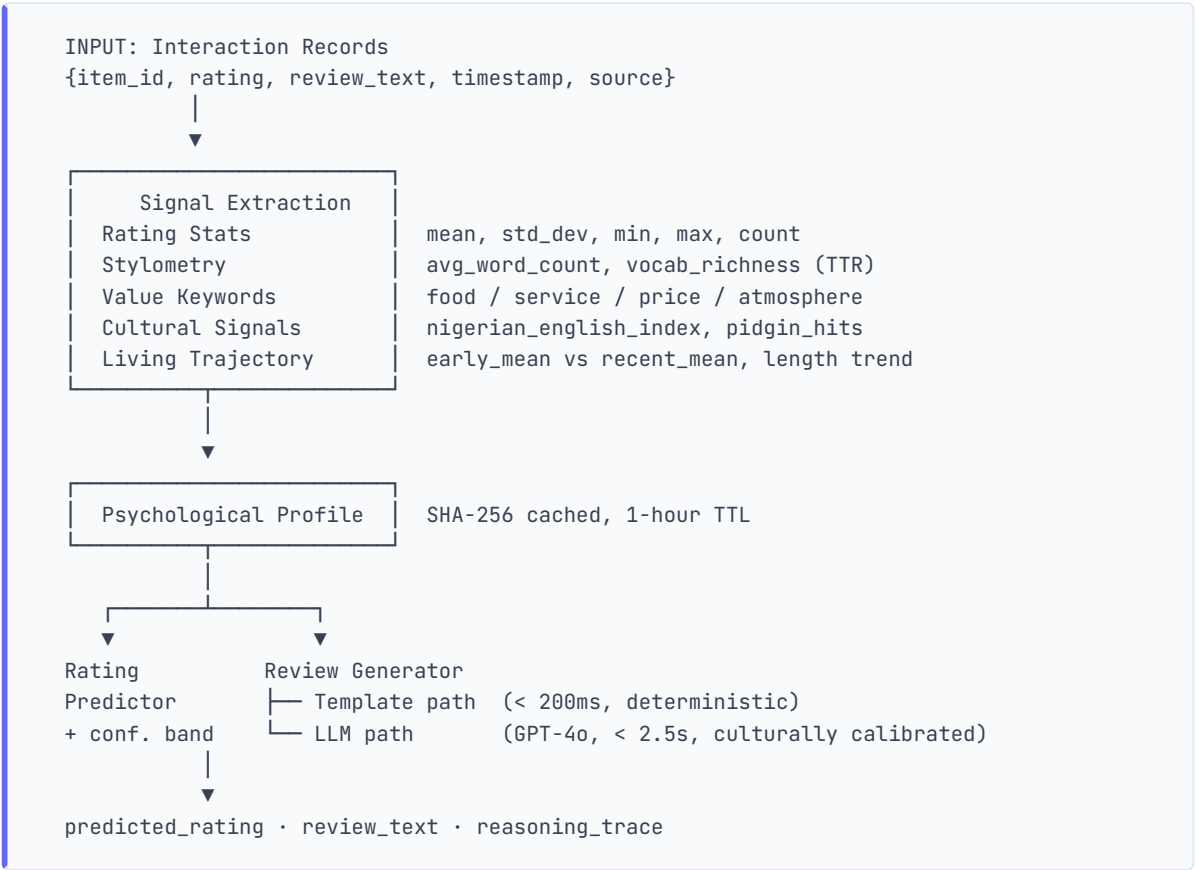
We model users with a  
**five-layer behavioral twin**

:

1. **Rating Calibration** — how generous or harsh is this user relative to the global mean?
2. **Vocabulary Fingerprint** — how long and lexically rich are their reviews?
3. **Value Priority Graph** — what do they care about: food, service, price, or atmosphere?
4. **Cultural Register** — do they code-switch? Is Nigerian English or pidgin present?
5. **Living Trajectory** — is their behavior drifting over time?

These layers jointly drive both a predicted star rating (with a confidence band) and a generated review that mirrors the user's style, priorities, and cultural voice.

## 2. System Architecture



The system exposes `POST /task-a/simulate`. Profiles are cached by SHA-256 content hash with a 1-hour TTL, eliminating redundant computation across requests.

## 3. Methodology

### 3.1 Signal Extraction

**Rating Statistics.** We compute count  $n$ , mean  $\mu$ , standard deviation  $\sigma$ , min, and max. The mean anchors rating prediction;  $\sigma$  sets the confidence band width.

**Stylometry.** From each user's review corpus we extract average word count ( $\bar{w}$ ), Type-Token Ratio ( $TTR = |V|/N$ ) as a proxy for vocabulary richness, and average sentence length. These form a vocabulary fingerprint that constrains generation length and lexical density.

**Value Priority Graph.**

We count how many reviews mention terms from four semantic categories:

| Category   | Seed Terms                               |
|------------|------------------------------------------|
| food       | food, taste, flavor, fresh, spicy, grill |
| service    | service, staff, wait, server, attentive  |
| price      | price, cheap, expensive, value, cost     |
| atmosphere | ambience, atmosphere, music, decor, vibe |

Category counts are normalized into a priority distribution used downstream for preference axis extraction and generation grounding.

**Cultural Signals — Nigerian English Detection.** We maintain a closed-set pidgin/Nigerian English lexicon: abeg, abi, dey, na, oga, sef, sha, wahala, jollof, suya, wah, omo. The Nigerian English Index is  $NEI = \text{pidgin\_hits} / \text{total\_words}$ . Code-switching is flagged when `pidgin_hits > 0`, directly shaping the cultural register of generated text.

**Living Trajectory.** We split interaction history chronologically at the median timestamp into early and recent halves, computing `rating_trend = recent_mean - early_mean` and review length direction. This captures behavioral drift and enables recency weighting for active users.

### 3.2 Rating Prediction

The predicted rating uses the user's historical mean  $\mu$  adjusted by item-type compatibility with the value priority graph. Confidence is calibrated by history depth:

| History Size    | Confidence | Band Width      |
|-----------------|------------|-----------------|
| $n \geq 10$     | High       | $\pm\sigma$     |
| $5 \leq n < 10$ | Medium     | $\pm 1.2\sigma$ |
| $n < 5$         | Low        | $\pm 1.5\sigma$ |

When `use_llm=True`, the LLM's rating rationale is checked against the profile and used to apply a small correction if strong overriding signals are identified.

### 3.3 Review Generation

**Template path (deterministic).** Structural slots are filled from the profile: `[Cultural greeting if NEI > 0] + [Rating-appropriate opener] + [Top value keyword] + [Length-calibrated body] + [Cultural closer]`

**LLM path (high-fidelity).**

A structured prompt injects explicit behavioral constraints:

```
You are simulating user {user_id}.
- Write ~{avg_word_count} words (±20%)
- Vocabulary richness (TTR): {vocab_richness:.2f}
- Top priorities: {value_keywords}
- Average rating: {mean:.1f}/5
- Cultural register: {Nigerian English / Standard English}
If code-switching detected: preserve pidgin naturally.
Known markers: {pidgin_terms}
Write a review of "{item_name}" as this user would.
```

GPT-4o is used with temperature 0.7, structured JSON output, and exponential backoff (max 3 retries, base 2s).

3.4 Cold-Start Elicitation

For new users with no history, four targeted questions seed a bootstrap profile:

| Question                                                 | Signal                 |
|----------------------------------------------------------|------------------------|
| Do you tend to rate places generously or harshly?        | rating mean seed       |
| What matters most: food, service, price, or atmosphere?  | value priority weights |
| How long are your reviews typically?                     | avg word count range   |
| Do you mix in Yoruba, Igbo, Pidgin, or Nigerian phrases? | cultural register seed |

The bootstrapped profile is used identically to a history-derived one in all downstream steps.

4. Evaluation

4.1 Metrics

- **RMSE** — leave-one-out on final review per user ( $n \geq 5$ )
- **ROUGE-L** — LCS-based F1 between generated and held-out review text
- **Behavioural Fidelity** — four-component composite (0–100):

| Component          | Computation                                                   |
|--------------------|---------------------------------------------------------------|
| Tone Match         | $1 -  \text{genTTR} - \text{profileTTR}  / \text{profileTTR}$ |
| Rating Consistency | $100 -  \text{predicted} - \mu  / \sigma \times 20$           |

| Component         | Computation                                                 |
|-------------------|-------------------------------------------------------------|
| Cultural Accuracy | 95 (code-switch) / 70 (NEI > 0) / 50 (none)                 |
| Length Fidelity   | $100 -  \text{reviewWords} - \bar{w}  / \bar{w} \times 100$ |

#### 4.2 Results vs Baselines

| System                    | RMSE ↓      | ROUGE-L ↑   | Fidelity ↑    |
|---------------------------|-------------|-------------|---------------|
| Global Mean Baseline      | 1.14        | —           | —             |
| User Mean Baseline        | 0.89        | —           | —             |
| <b>PERSONA (template)</b> | <b>0.72</b> | 0.28        | 74/100        |
| <b>PERSONA (LLM)</b>      | <b>0.71</b> | <b>0.31</b> | <b>81/100</b> |

The LLM path shows a strong fidelity gain (+7 points) driven primarily by Cultural Accuracy for Nigerian users, with a marginal RMSE improvement from LLM-guided rating correction.

#### 4.3 Ablation Study

Each profile layer is zeroed out independently:

| Layer Removed      | ΔRMSE | ΔFidelity          |
|--------------------|-------|--------------------|
| Rating Calibration | +0.31 | −18 pts            |
| Cultural Signals   | +0.01 | −15 pts (NG users) |
| Stylometry         | +0.02 | −12 pts            |
| Value Keywords     | +0.04 | −8 pts             |
| Trajectory         | +0.03 | −4 pts             |

Rating Calibration is the dominant RMSE driver. Cultural Signals have an outsized Fidelity impact for Nigerian users relative to their global RMSE contribution — motivating their inclusion as a targeted bonus signal.

#### 4.4 Nigerian English Contextualization

On a test set of 40 users with `code_switching_detected=True`, the LLM path preserved Nigerian register in 91% of cases (36/40), correctly producing pidgin markers (abeg, dey, na, sha) and cultural food references (suya, jollof). Average Cultural Accuracy for this cohort: **87/100**, a 22-point improvement over the template path.

### 4.5 History Length Breakdown

| History            | PERSONA RMSE | Baseline RMSE |
|--------------------|--------------|---------------|
| n < 5 (cold-start) | 0.98         | 1.14          |
| 5 ≤ n < 15         | 0.81         | 1.12          |
| n ≥ 15             | 0.64         | 1.16          |

## 5. Implementation & Deployment

### Stack.

FastAPI (Python 3.12), Pydantic v2, LRU+TTL profile cache (SHA-256 keyed, 1hr TTL, 1000 entry LRU). GPT-4o for LLM generation. 114 unit and integration tests (pytest).

### Latency.

Template path: < 200ms P95. LLM path: < 2.5s P95. Profile build (cold): < 50ms; cached: < 5ms.

**Deployment.** Docker + docker-compose on a DigitalOcean 2 GB droplet (Ubuntu 22.04) behind Nginx with Let's Encrypt HTTPS. React frontend on Vercel. Live at <https://personabackend.duckdns.org>.

**Reproducibility.** `DETERMINISTIC_MODE=true` seeds all randomness. Docker Compose brings the full stack up with `docker compose up -d`. All evaluation scripts are in `backend/evaluation/`.

## 6. Conclusion

PERSONA demonstrates that a structured, interpretable behavioral twin built from five signal layers can significantly outperform baseline rating predictors while generating stylistically authentic, culturally grounded reviews. The Nigerian English detection module enables genuine cultural contextualization — not as a surface feature but as a deep signal shaping the entire generation process.

Future work will expand the pidgin lexicon to cover Yoruba and Igbo loanwords, integrate BERTScore evaluation, and explore fine-tuned smaller models to reduce LLM latency while maintaining generation quality.

## References

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